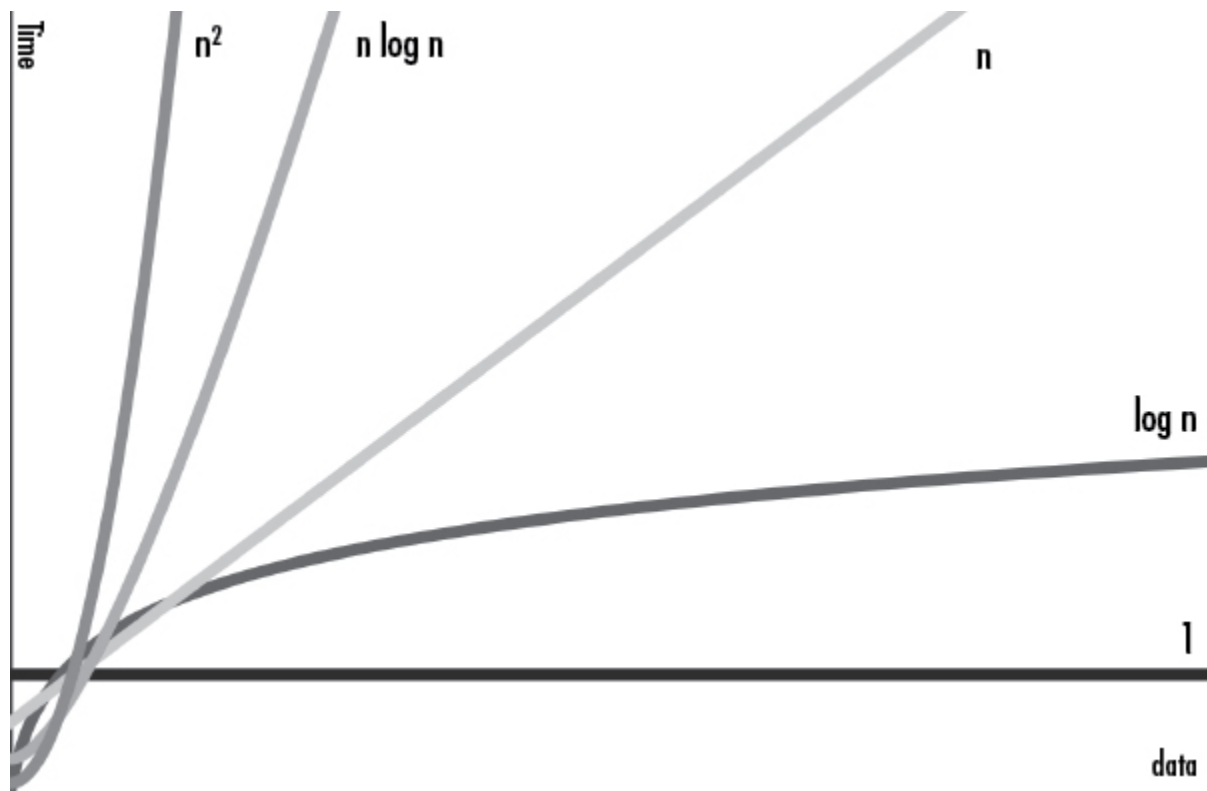




[Figure 13-3](#): The graph of the big O orders

As you can see, the runtime of the higher orders grows at a faster rate than that of the lower orders. Although the lower orders could have large coefficients that make them temporarily larger than the higher orders, the higher orders eventually outpace them.

Big O Analysis Examples

Let's determine the big O orders of some example functions. In these examples, we'll use a parameter named `books` that is a list of strings of book titles.

The `countBookPoints()` function calculates a score based on the number of `books` in the `books` list. Most books are worth one point, and books by a certain author are worth two points:

```
def countBookPoints(books):
    points = 0          # 1 step
    for book in books:  # n * steps in the loop
        points += 1     # 1 step
```